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Research topic: The Effects of Short-Form Video Exposure on Attentional Control and Working Memory in Adolescents: Investigating the Role of Dopaminergic Reward Processing

The Effects of Short-Form Video Exposure on Attentional Control and Working Memory in Adolescents: Investigating the Role of Dopaminergic Reward Processing

Introduction

Short-form video usually demonstrates viewers with quick, engaging clips, and rapid content to shift between topics, usually shown vertically and the duration is under 60 seconds [1]. This format raises an important question about how attention is allocated during viewing and whether its effects extend to subsequent cognitive tasks. For adolescents, this question is particularly relevant to activities requiring sustained concentration and the retention of information. Attentional control refers to the ability to direct attention toward relevant information while resisting distractions [2]. Working memory involves temporarily maintaining, restoring, and manipulating information [3]. Investigating these distinct but related functions could clarify whether short-form video exposure affects processes needed for following instructions, reading, and problem-solving.

Dopaminergic Reward Process refers to the role of dopamine in how the brain respond to and learn from rewarding stimuli. Dopamine neurons are particularly involved in reward prediction error, which occurs when an actual reward differs from what was expected [4]. Dopamine is also associated with the motivational value or "wanting" of rewarding stimuli rather than simply producing pleasure [5]. Therefore, repeated exposure to rapidly changing and potentially rewarding stimuli, such as short-form videos, may interact with reward-related processes that influence attention and behavior.

Research questions

How does 15 minutes of short-form video exposure, compared with 15 minutes of continuous video viewing, affect attentional control and working memory performance in adolescents aged 15–18, and to what extent does individual reward sensitivity moderate these effects?

This study will address three sub-questions:

1. Does short-form video exposure produce a greater change in attentional control than continuous video viewing?
2. Does short-form video exposure lead to a greater change in working memory performance than continuous video viewing?
3. Does baseline reward sensitivity predict the magnitude of these effects?

Research aim

This study aims to investigate the immediate effects of 15 minutes of short-form video exposure on attentional control and working memory in adolescents aged 15–18, compared with an equal duration of continuous video viewing. It will also examine whether individual differences in baseline reward sensitivity moderate changes in cognitive performance.

Existing Literature

Existing research provides several complementary perspectives on short-form video use and cognition, including experimental evidence on memory, observational research involving adolescents, and neuroimaging studies of reward processing. However, these studies examine different cognitive outcomes and viewing conditions. Their findings therefore need to be distinguished carefully when evaluating possible effects on attentional control and working memory [6][7][8].

Chiossi et al. [6] investigated the effects of different social media activities on prospective memory in a between-subjects experiment involving 60 participants. Participants were required to remember and execute previously formed intentions after interruptions involving TikTok, Twitter, YouTube, or no activity. The TikTok condition produced significantly poorer prospective memory performance. This supports the possibility that short videos combined with rapid context switching disrupt the retention of task goals. Nevertheless, prospective memory concerns remembering intended actions, whereas working memory involves maintaining and manipulating information during an ongoing task.

Reward processing provides possible explanations for why short-form video can capture people's attention. Using fMRI, Su et al. [7] found greater activation in the ventral tegmental area and components of the default mode network during personalized compared with non-personalized video viewing. These findings provide a rationale for considering reward-related processes, but they do not demonstrate that dopamine causes subsequent attention or memory difficulties. The study also compared personalization conditions rather than short-form and continuous video formats. Its relevance to the proposed study is therefore mechanistic and indirect.

Research involving adolescents provides more direct evidence concerning working memory. Xu et al. [8] examined short-video application use, cognitive functioning, and academic performance in younger and older adolescent cohorts in China. Their work identified age-related differences in the associations: reliance on algorithmic recommendations was negatively associated with working memory and other outcomes in the younger cohort, but the same pattern was not established in the older cohort. This challenges the assumption that adolescents form a uniformly susceptible group. However, the observational design cannot establish whether short-video use reduces cognitive performance or whether pre-existing cognitive differences influence viewing habits. These limitations support experimentally examining immediate changes in the proposed 15–18 age group. [8]

Taken together, these existing studies support further research on short-form video exposure, but its effects may differ between individuals and may not always be caused by dopamine. This proposal will extend its evidence by measuring changes in attentional control and working memory while maintain equal duration of video. It will also explore the baseline reward sensitivity as a moderator.

Methodology

1. Research design:

This study will use randomized, pre-test – post-test experimental design to investigate the immediate effects of short-form video exposure on adolescents' cognitive performance. Participants will be assigned to either a short-form video condition or a continuous video condition. Both groups will watch videos for 15 minutes and complete cognitive tasks before and after viewing. The cognitive tasks will be 2-back task and computerized color-word Stroop tests.

The independent variable will be video format. The dependent variables will be attentional control, measured through Stroop interference, and working memory performance, measured through a computerized 2-back task. Baseline reward sensitivity will be examined as a possible moderator. Dopaminergic reward processing will provide a theoretical framework, but dopamine activity will not be directly measured.

2. Sampling method:

The proposed recruitment target will be 128 adolescents aged 15–18, with approximately 64 participants in each condition. Participants will be recruited voluntarily through participating secondary schools. This target is a preliminary planning figure. An a priori power analysis will be completed before recruitment to confirm the sample required for the primary comparisons. Analyses of reward sensitivity will be treated as exploratory because interaction effects may require a larger sample. Convenience recruitment will limit how broadly the findings can be generalized.

3. Measurement:

Attentional control will be measured using a computerized Stroop task [9]. Participants will identify the color of words while ignoring their meaning. Greater Stroop interference will indicate weaker attentional control.

Working memory will be measured using a computerized 2-back task, in which participants identify whether a stimulus matches one shown two positions earlier. Higher performance will indicate better working memory.

Reward sensitivity will be measured using the BAS Reward Responsiveness subscale from Carver and White's BIS/BAS scales [10]. Scores will represent self-reported reward responsiveness rather than dopamine levels.

4. Data analysis:

Post-test Stroop and 2-back performance will be compared between the two video conditions while controlling for baseline scores.

Reward sensitivity will also be tested as a possible moderator to examine whether the effects of short-form video exposure differ between participants with different levels

of reward responsiveness.

Effect sizes, confidence intervals, and p-values will be reported. Age, sleep duration, and habitual short-form video use may also be considered as control variables.

Ethical considerations

The study will begin only after the required school and research-ethics approvals have been obtained. Participants under 18 will provide assent alongside parental or guardian consent; participants aged 18 will provide their own informed consent.

The debriefing will explain that the study investigates short-term performance differences and cannot establish lasting cognitive harm, diagnose an attention disorder, or confirm a dopaminergic mechanism.

Potential Limitations

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This study has several possible limitations. First, participants recruited from selected schools may not represent all adolescents. Second, 15 minutes of exposure can only reveal immediate effects, not the long-term effects of regular viewing. Third, the prepared videos will not include personalized recommendations or scrolling, making the experience different from everyday social-media use. Differences in narrative coherence may also contribute to the results.

Conclusion

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In conclusion, this proposed study aims to examine whether 15 minutes of short-form video exposure affects attentional control and working memory in adolescents aged 15–18 compared with continuous video viewing. The findings could contribute to understanding how video format influences adolescents' performance on tasks relevant to learning. However, the study will not establish long-term effects or directly confirm the role of dopamine. Instead, it will provide behavioral evidence that could guide future research into the relationship between digital media, reward processing, and cognitive control.

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